

Enhancing Risk Management Effectiveness in Hybrid Projects through AI-Driven Strategic Governance

A Structural Equation Modelling Study

Based on primary data from 384 project professionals

Analysis: Structural Equation Modelling (SEM) | SPSS & AMOS

Introduction & Research Context

The Problem

Traditional risk management relies on static registers, periodic assessments, and subjective expert judgment — insufficient for today's complex, data-intensive projects.

Hybrid Complexity

Agile + Waterfall coexistence creates fragmented governance, inconsistent risk documentation, and ambiguous escalation authority.

The AI Opportunity

AI enables predictive analytics, real-time monitoring, continuous learning, and data-driven decision support for risk governance.

Research Gap: No integrated SEM model testing AI capability alongside governance mechanisms in hybrid project risk management

Research Objectives & Hypotheses

H1

Risk Process Alignment → Risk Management Effectiveness

H2

Risk Visibility Integration → Risk Management Effectiveness

H3

Timely Risk Escalation → Risk Management Effectiveness

H4

Governance Consistency → Risk Management Effectiveness

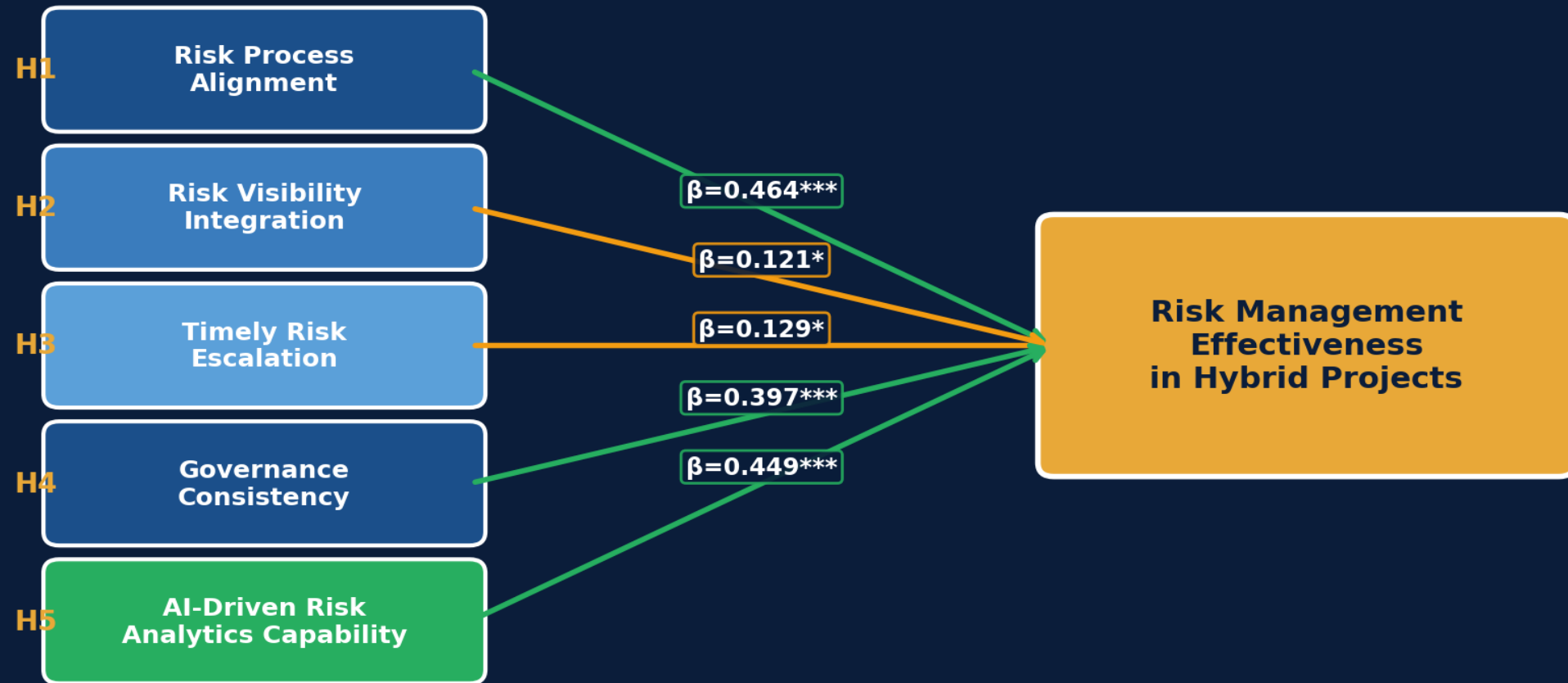
H5

AI-Driven Risk Analytics Capability → Risk Management Effectiveness

All hypotheses posit a significant positive impact on Risk Management Effectiveness in Hybrid Project Environments.

Conceptual Framework

Integrated Model of AI-Driven Strategic Risk Management



■ Strong significance ($p < 0.001$)

■ Moderate significance ($p < 0.05$)

Research Methodology

Quantitative Research Design

Cross-sectional survey
with 5-point Likert scale

384 Professionals Surveyed

Project managers, risk managers,
PMO, analysts, scrum masters

SPSS + AMOS Analysis Tools

CFA for measurement model
SEM for hypothesis testing

Rigorous Validation

KMO = 0.927 | All AVE > 0.5
All CR > 0.7 | Fornell-Larcker

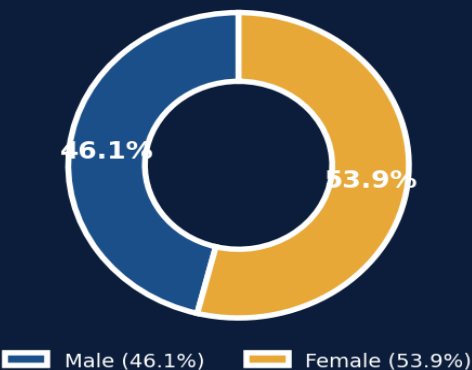
Statistical Techniques

- Confirmatory Factor Analysis (CFA) for measurement model validation
- Structural Equation Modelling (SEM) for testing causal relationships
- Discriminant Validity via Fornell-Larcker Criterion | Cronbach's Alpha for reliability

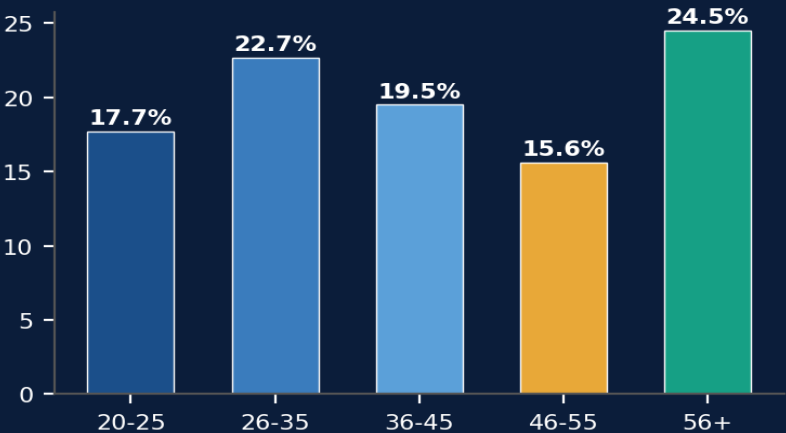
Sample Demographics

N = 384 Project Professionals across Diverse Industries

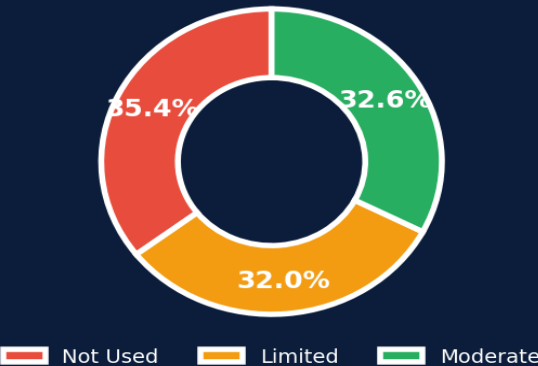
Gender



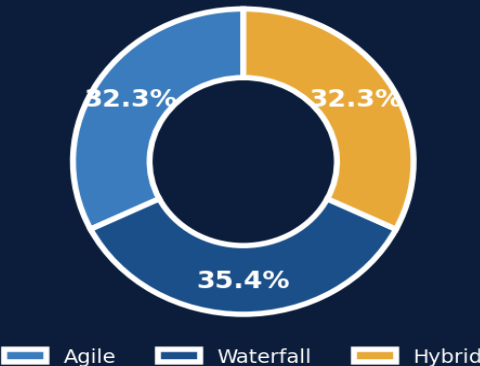
Age Group (%)



AI Usage in Risk Mgmt



Primary Methodology



77%+ Over 5 years experience

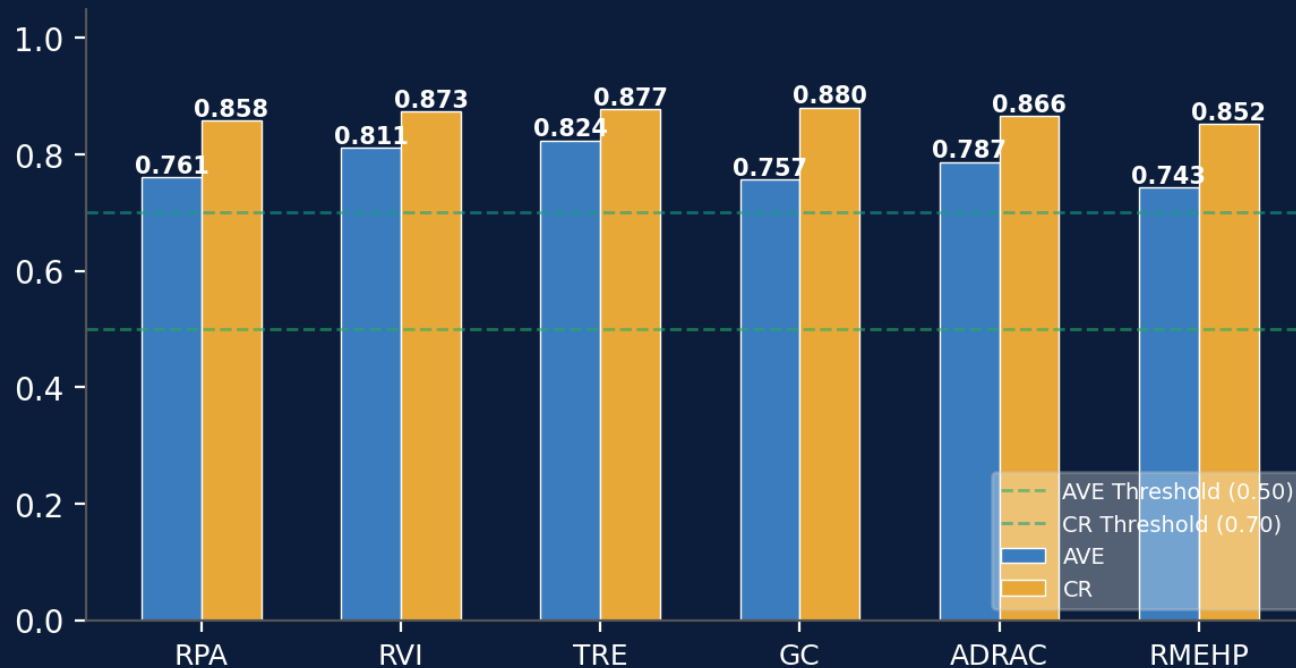
53.9% Female respondents

27.6% Hold professional certifications

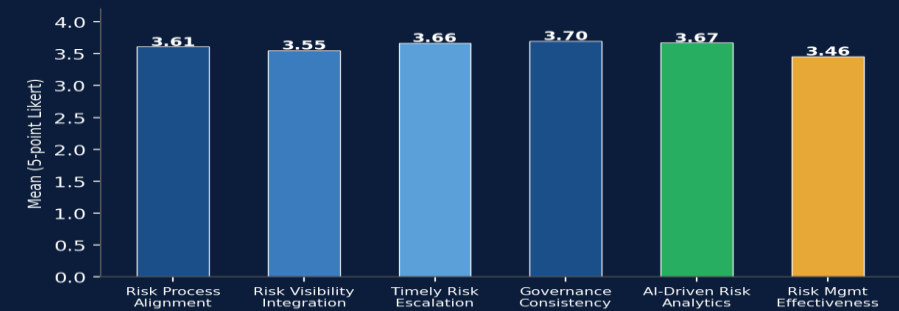
32.3% Hybrid project experience

Measurement Model & Validity

Convergent Validity: AVE & Composite Reliability



Mean Scores of Study Variables



Measurement Validation Summary

KMO = 0.927

Excellent sampling adequacy

All AVE > 0.50

Convergent validity confirmed

All CR > 0.70

Internal consistency established

Fornell-Larcker Criterion Met

Discriminant validity verified

Bartlett's Test: $p < 0.001$

$\chi^2 = 7618.445$, $df = 465$

All factor loadings > 0.60

Indicator reliability confirmed

Hypothesis Results: H1 & H2

H1: Risk Process Alignment → RME

✓ SUPPORTED

$\beta = 0.464$ | C.R. = 7.346 | $p < 0.001$

Model Fit:

CMIN/DF = 2.366 | GFI = 0.958 | CFI = 0.976

RMSEA = 0.060

Interpretation:

Moderate-to-strong positive effect. Aligning risk identification, assessment, and monitoring across Agile & Waterfall components significantly enhances coordination and reduces ambiguity.

Process alignment functions as a foundational enabler of effective risk management.

H2: Risk Visibility Integration → RME

⚠ SUPPORTED (Marginal Effect)

$\beta = 0.121$ | C.R. = 2.116 | $p = 0.034$

Model Fit:

CMIN/DF = 1.376 | CFI = 0.994

RMSEA = 0.031

Interpretation:

Statistically significant but modest effect. Visibility alone is insufficient — it functions as a supportive mechanism rather than an independent driver.

Without strong governance and interpretive capacity, information availability does not translate into better outcomes.

Hypothesis Results: H3 & H4

H3: Timely Risk Escalation → RME

⚠️ **SUPPORTED (Modest Effect)**

$\beta = 0.129$ | C.R. = 2.283 | $p = 0.022$

Model Fit:

CMIN/DF = 1.326 | CFI = 0.995

RMSEA = 0.029

Interpretation:

Positive but modest direct effect. In hybrid settings, Agile teams manage risks locally, reducing the need for frequent formal escalation.

Escalation effectiveness depends on appropriateness and quality rather than on speed alone.

H4: Governance Consistency → RME

✅ **SUPPORTED**

$\beta = 0.397$ | C.R. = 6.869 | $p < 0.001$

Model Fit:

CMIN/DF = 1.716 | CFI = 0.985

RMSEA = 0.043

Interpretation:

Governance Consistency is the structural backbone that stabilizes risk management across diverse execution approaches.

Uniform application of policies, decision rights, and accountability mechanisms reduces ambiguity and enhances coordination.

Hypothesis 5 – Key Finding

AI-Driven Risk Analytics Capability → Risk Management Effectiveness



H5 STRONGLY SUPPORTED

$\beta = 0.449$ | C.R. = 7.697 | $p < 0.001$

Model Fit:

CMIN/DF = 1.995 | NFI = 0.965 | CFI = 0.982 | RMSEA = 0.051

Key Insight:

AI-Driven Risk Analytics is one of the STRONGEST predictors of Risk Management Effectiveness among all variables tested.

AI functions as an organizational capability that enhances predictive foresight, anomaly detection, real-time monitoring, and automated risk prioritization.

It complements and amplifies governance structures rather than replacing them.

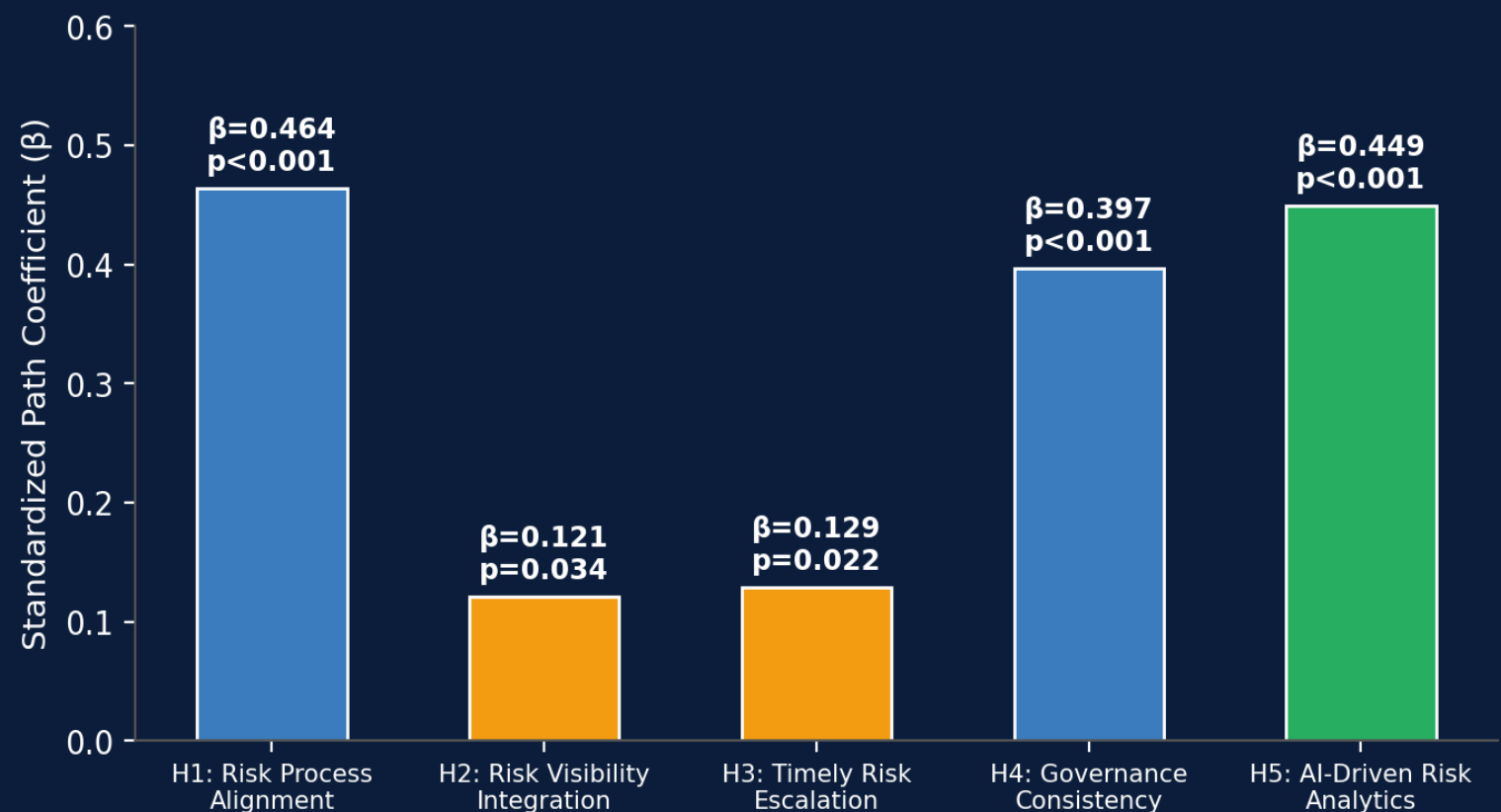
**Strongest Predictor
Among All 5 Variables**

**Strategic Capability
Not Just a Tech Tool**

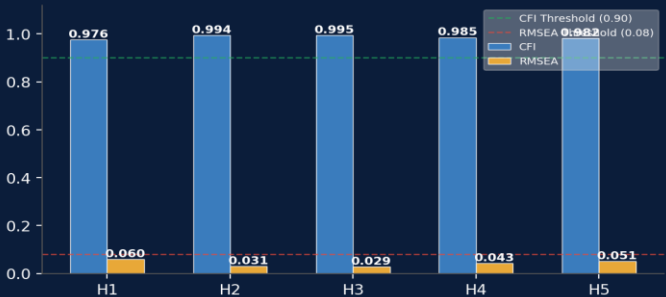
**Amplifies Governance
& Decision Quality**

Summary of All Hypothesis Results

SEM Path Coefficients - All Hypotheses



Model Fit Indices Across Hypotheses



All 5 models exceed fit thresholds

- ✓ CFI > 0.90
- ✓ RMSEA < 0.08
- ✓ CMIN/DF < 3.0

Outcome Summary

H	Path	β	Result
H1	RPA $\rightarrow$ RME	0.464	Supported
H2	RVI $\rightarrow$ RME	0.121	Marginal
H3	TRE $\rightarrow$ RME	0.129	Modest
H4	GC $\rightarrow$ RME	0.397	Supported
H5	ADRAC $\rightarrow$ RME	0.449	Supported

Rank: H5 (AI) > H1 (RPA) > H4 (GC) >> H3 > H2

Discussion & Key Insights

Structural Coherence Drives Effectiveness

Governance Consistency and Risk Process Alignment are foundational enablers. Risk management effectiveness requires deliberate integration across Agile and Waterfall governance — not parallel or fragmented systems.

AI as a Strategic Organizational Capability

AI analytics is not merely a technical tool but an organizational capability that enhances predictive foresight and decision quality. It amplifies governance structures and reduces cognitive limitations in complex environments.

Visibility & Speed Are Necessary but Insufficient

Risk visibility and timely escalation are supportive but not dominant drivers. Without governance clarity and interpretive capacity, information availability alone does not improve risk outcomes. Quality of decisions matters more than speed.

Theoretical & Practical Implications

Theoretical Contributions

- Positions AI as a structural determinant of risk management effectiveness, not just a peripheral tool
-
- Advances hybrid project management research through integrated SEM-based empirical validation
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- Bridges governance theory and dynamic capability theory in the hybrid project domain
-
- Demonstrates AI capability as complementary to governance structures, not a replacement for them

Practical Recommendations

- Prioritize harmonizing risk processes across Agile and Waterfall components
-
- Establish consistent governance: clear decision rights, accountability, and escalation pathways
-
- Invest in AI analytics embedded within governance frameworks — not deployed in isolation
-
- Train managers to interpret and act on AI-generated insights for strategic decisions

Conclusion, Limitations & Future Research

Effective risk management in hybrid projects requires integrated structural alignment, governance stability, and AI-driven analytical intelligence — not isolated practices or technologies.

①

Governance + Process Alignment

Structural coherence is the foundation for effective risk governance.

②

AI as Strategic Enabler

Strongest predictor — enhances foresight and amplifies governance.

③

Beyond Visibility & Speed

Decision quality and governance clarity matter more than information volume.

Limitations

- Cross-sectional design limits causal inference
- India-focused sample — may limit generalizability
- Self-reported data — potential response bias

Future Research

- Longitudinal studies across project phases
- Cross-cultural replication in other regions
- Mediating roles of culture & project complexity

Thank You — Questions & Discussion Welcome